

Cenk Baykal

ML & QUANT RESEARCHER

baykal@alum.mit.edu | [baykalc.github.io](https://github.com/baykalc) | linkedin.com/in/cenkbaykal | [Google Scholar](https://scholar.google.com/citations?user=...)

EDUCATION

Massachusetts Institute of Technology

PH.D. IN COMPUTER SCIENCE (GPA: 5.00/5.00)

- Minor: Probability in High Dimension (Mathematics)
- Thesis: *Sampling-based Algorithms for Fast and Deployable AI*
- Advisor: Daniela Rus

Cambridge, MA

2017–2021

Massachusetts Institute of Technology

S.M. IN COMPUTER SCIENCE (GPA: 4.91/5.00)

- Thesis: *Algorithms for Persistent Autonomy and Surveillance*
- Advisor: Daniela Rus

Cambridge, MA

2015–2017

University of North Carolina at Chapel Hill

B.S. COMPUTER SCIENCE WITH HIGHEST HONORS, B.A. MATHEMATICS (GPA: 3.91/4.00)

- Graduated with Highest Distinction
- Thesis: *Design Optimization Algorithms for Concentric Tube Robots*
- Advisor: Ron Alterovitz

Chapel Hill, NC

2011–2015

EXPERIENCE

Two Sigma

QUANTITATIVE RESEARCHER

- Designed systematic alpha strategies that leverage alternative data sources and advanced machine learning techniques
- Built and maintained research tooling using agentic AI and LLMs to accelerate signal capture and forecast evaluation

New York City, NY

July 2024–Present

Google Research

RESEARCH SCIENTIST

- Developed conditional computation algorithms for transformers that enabled up to 30% speedups on large language models (LLMs)
- Designed data-efficient knowledge distillation strategies that led to improved transformer architectures with only 50% of teacher labeling cost
- Mentored scholars in Google CSRMP, supporting project design and research execution for students from historically marginalized groups
- Received the 2023 Google Research Tech Impact Award for leading “high-impact projects made sustainable by achieving tech excellence and great team dynamics”
- Received a Google Spot Bonus for “critical contributions to the efficiency of compact Gemini models”

Cambridge, MA

January 2022–June 2024

Massachusetts Institute of Technology

POSTDOCTORAL RESEARCHER

- Worked on algorithms for privacy-aware and efficient machine learning

J.P. Morgan AI Research

AI RESEARCH INTERN

- Developed sampling-based algorithms with regret guarantees for large-scale graph neural network training

Massachusetts Institute of Technology

TEACHING ASSISTANT FOR ADVANCED ALGORITHMS (6.854J / 18.415J)

- Conducted office hours to help students on problem sets and concepts covered in lectures; designed and graded assignments
- TA rating according to the official MIT subject evaluation report: 7.0/7.0

Cambridge, MA

Fall 2019

Microsoft, SAS Institute, University of North Carolina at Chapel Hill

EARLIER INDUSTRY & TEACHING EXPERIENCE

- Shipped SQL Server compression optimizations and analytics tooling deployed across academic and enterprise users
- Conducted research in robotics, path planning, and enabling technologies

Redmond, WA; Cary, NC;

Chapel Hill, NC

2012–2015

HONORS & AWARDS

2024	Google Spot Bonus , For critical contributions to the efficiency of compact Gemini models	Google
2023	Google Research Tech Impact Award , For contributions to ML efficiency techniques	Google
2023	NeurIPS Spotlight Paper , Alternating Updates for Efficient Transformers	NeurIPS
2021	Winner , MIT The Engine's Interval Program (one of two winning teams)	MIT
2020–2021	Expert Reviewer , ICLR (2020, 2021); ICML (2021)	Program Committees
2020	Top 10% Reviewer , NeurIPS	Program Committee
2017	RSS Best Paper Award , Robotics: Science and Systems Conference	RSS
	Early Academic Honors , UNC distinctions including Carolina Research Scholar, CRA Outstanding	University of North
2011–2015	Undergraduate Finalist, Phi Beta Kappa, SAS Charles H. Dunham Scholarship, Dunlevie Honors Award, Summer Undergraduate Research Fellowship, Honors Carolina, Dean's List	Carolina at Chapel Hill

SELECTED PUBLICATIONS

For the complete list of publications, visit [Google Scholar](#).

TRANSFORMERS & DISTILLATION

Alternating Updates for Efficient Transformers

CENK BAYKAL, DYLAN CUTLER, NISHANTH DIKKALA, NIKHIL GHOSH, RINA PANIGRAHY, XIN WANG

[NeurIPS \(Spotlight\)](#)

2023

SLaM: Student-Label Mixing for Distillation with Unlabeled Examples

VASILIS KONTONIS, FOTIS ILIOPOULOS, KHOA TRINH, CENK BAYKAL, GAURAV MENGHANI, ERIK VEE

[NeurIPS](#)

2023

Robust Active Distillation

CENK BAYKAL, KHOA TRINH, FOTIS ILIOPOULOS, GAURAV MENGHANI, ERIK VEE

[ICLR](#)

2023

A Theoretical View on Sparsely Activated Networks

CENK BAYKAL, NISHANTH DIKKALA, RINA PANIGRAHY, CYRUS RASHTCHIAN, XIN WANG

[NeurIPS](#)

2022

COMPRESSION & ROBOTICS

SiPPing Neural Networks: Sensitivity-informed Provable Pruning of Neural Networks

CENK BAYKAL*, LUCAS LIEBENWEIN*, IGOR GILITSCHENSKI, DAN FELDMAN, DANIELA RUS

[SIAM SIMODS](#)

2022

Lost in Pruning: The Effects of Pruning Neural Networks beyond Test Accuracy

LUCAS LIEBENWEIN, CENK BAYKAL, BRANDON CARTER, DAVID GIFFORD, DANIELA RUS

[MLSys](#)

2021

Provable Filter Pruning for Efficient Neural Networks

CENK BAYKAL*, LUCAS LIEBENWEIN*, HARRY LANG, DAN FELDMAN, DANIELA RUS

[ICLR](#)

2020

Data-Dependent Coresets for Compressing Neural Networks with Applications to Generalization Bounds

CENK BAYKAL*, LUCAS LIEBENWEIN*, IGOR GILITSCHENSKI, DAN FELDMAN, DANIELA RUS

[ICLR](#)

2019

Sampling-Based Approximation Algorithms for Reachability Analysis with Provable Guarantees

CENK BAYKAL*, LUCAS LIEBENWEIN*, IGOR GILITSCHENSKI, SERTAC KARAMAN, DANIELA RUS

[Robotics: Science and Systems](#)

2018

Asymptotically Optimal Design of Piecewise Cylindrical Robots using Motion Planning

CENK BAYKAL, RON ALTEROVITZ

[Robotics: Science and Systems \(Best Paper Award\)](#)

2017